ETSI GS QKD 004 push mode specification, QKD server v1.0.0

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> Scroll down for code samples, example requests and responses. Select a language for code samples from the tabs above or the mobile navigation menu.

Server description of the QKD endpoint processing ETSI GS QKD 004 push mode requests

Base URLs:

- https://qkd_server/api/v1/kms/etsi004

Web: Support

etsi004

post_open_connect

> Code samples

```
# You can also use wget
curl -X POST https://qkd_server/api/v1/kms/etsi004/open-connect \
  -H 'Content-Type: application/json' \
  -H 'Accept: application/json'
```

```
POST https://qkd_server/api/v1/kms/etsi004/open-connect HTTP/1.1
Host: qkd_server
Content-Type: application/json
Accept: application/json
```

```
const inputBody = '{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
```

```
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  }
}';
const headers = {
  'Content-Type':'application/json',
  'Accept':'application/json'
};

fetch('https://qkd_server/api/v1/kms/etsi004/open-connect',
{
  method: 'POST',
  body: inputBody,
  headers: headers
})
.then(function(res) {
    return res.json();
}).then(function(body) {
    console.log(body);
});
```

```
require 'rest-client'
require 'json'

headers = {
  'Content-Type' => 'application/json',
  'Accept' => 'application/json'
}

result = RestClient.post 'https://qkd_server/api/v1/kms/etsi004/open-connect',
  params: {
  }, headers: headers

p JSON.parse(result)
```

```
import requests
headers = {
  'Content-Type': 'application/json',
  'Accept': 'application/json'
}

r = requests.post('https://qkd_server/api/v1/kms/etsi004/open-connect', headers = headers)
```

```python
print(r.json())
```

```php
<?php

require 'vendor/autoload.php';

$headers = array(
    'Content-Type' => 'application/json',
    'Accept' => 'application/json',
);

$client = new \GuzzleHttp\Client();

// Define array of request body.
$request_body = array();

try {
    $response = $client->request('POST','https://qkd_server/api/v1/kms/etsi004/open-connect'
        'headers' => $headers,
        'json' => $request_body,
       )
    );
    print_r($response->getBody()->getContents());
 }
 catch (\GuzzleHttp\Exception\BadResponseException $e) {
    // handle exception or api errors.
    print_r($e->getMessage());
 }

 // ...
```

```java
URL obj = new URL("https://qkd_server/api/v1/kms/etsi004/open-connect");
HttpURLConnection con = (HttpURLConnection) obj.openConnection();
con.setRequestMethod("POST");
int responseCode = con.getResponseCode();
BufferedReader in = new BufferedReader(
    new InputStreamReader(con.getInputStream()));
String inputLine;
StringBuffer response = new StringBuffer();
while ((inputLine = in.readLine()) != null) {
    response.append(inputLine);
}
in.close();
System.out.println(response.toString());
```

```go
package main
```

```go
import (
	"bytes"
	"net/http"
)

func main() {

    headers := map[string][]string{
        "Content-Type": []string{"application/json"},
        "Accept": []string{"application/json"},
    }

    data := bytes.NewBuffer([]byte{jsonReq})
    req, err := http.NewRequest("POST", "https://qkd_server/api/v1/kms/etsi004/open-connect"
    req.Header = headers

    client := &http.Client{}
    resp, err := client.Do(req)
    // ...
}
```

`POST /open-connect`

open key stream

The `open-connect` message connects a KMS to the QKD Device. Conforming to the specification, the QKD device will assign the `key_stream_id` and can suggest alternative QoS due to its capabilities.

Body parameter

```json
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  }
}
```

Parameters

Name	In	Type	Required	Description
body	body	object	true	none
» source	body	string(uri)	true	the source URI of the request.
» destination	body	string(uri)	true	the destination URI of the request.
» key_stream_id	body	string(uuid)	false	key stream ID. If not specified it is up to the QKD device to define the UUID
» qos	body	qos	true	All values according to ETSI GS QKD 004 specification. If KMS does not want to specify a parameter, it is up to the choice of the QKD. Either the value is given as “null” or not present.
»» key_chunk_size	body	integer(uint32)	false	length of key chunks in Byte.
»» max_bps	body	integer(uint32)	false	in bit per second
»» min_bps	body	integer(uint32)	false	in bit per second
»» jitter	body	integer(uint32)	false	in bit per second
»» priority	body	integer(uint32)	false	Meaning up to implementation.

Name	In	Type	Required	Description
»» timeout	body	integer(uint32)	false	Timeout in msec after which the call will be aborted, returning an error.
»» ttl	body	integer(uint32)	false	Time in seconds after which the keys for this key stream ID shall be erased from the kms key store.
»» metadata_mimetype	body	string(json)	false	MIME type of the metadata of the `get` request

Example responses

201 Response

```json
{
  "source": "http://kms-1.net",
  "destination": "http://kms-2.net",
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "qos": {
    "key_chunk_size": 32,
    "max_bps": 128,
    "min_bps": 64,
    "jitter": 1,
    "priority": 3,
    "timeout": 50,
    "ttl": 600,
    "metadata_mimetype": "application/json"
  },
  "status": 0
}
```

Responses

Status	Meaning	Description	Schema
201	Created	Key stream successfully created	Inline
400	Bad Request	Client sent a malformed input	None
500	Internal Server Error	The server experienced an internal error	None

Response Schema

Status Code **201**

Name	Type	Required	Restrictions	Description
» source	string(uri)	true	none	the source URI of the request.
» destination	string(uri)	true	none	the destination URI of the request.
» key_stream_id	string(uuid)	false	none	key stream ID. If not specified it is up to the QKD device to define the UUID

Name	Type	Required	Restrictions	Description
» qos	qos	true	none	All values according to ETSI GS QKD 004 specification. If KMS does not want to specify a parameter, it is up to the choice of the QKD. Either the value is given as “null” or not present.
»» key_chunk_size	integer(uint32)	false	none	length of key chunks in Byte.
»» max_bps	integer(uint32)	false	none	in bit per second
»» min_bps	integer(uint32)	false	none	in bit per second
»» jitter	integer(uint32)	false	none	in bit per second
»» priority	integer(uint32)	false	none	Meaning up to implementation.
»» timeout	integer(uint32)	false	none	Timeout in msec after which the call will be aborted, returning an error.
»» ttl	integer(uint32)	false	none	Time in seconds after which the keys for this key stream ID shall be erased from the kms key store.

Name	Type	Required	Restrictions	Description
»» metadata_mimetype	string(json)	false	none	MIME type of the metadata of the `get` request
» status	status(uint32)	false	none	none

oneOf

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful connection, but peer not connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because insufficient key available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because peer application is not yet connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	No QKD connection available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN_CONNECT failed because the KSID is already in use

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	TIMEOUT_ERROR The call failed because the specified TIMEOUT

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because metadata field size insufficient. Returned Metadata_size value holds minimum needed size of metadata

This operation does not require authentication

get_key

Code samples

```
# You can also use wget
curl -X GET https://qkd_server/api/v1/kms/etsi004/get-key \
  -H 'Content-Type: application/json' \
  -H 'Accept: application/json'
```

```
GET https://qkd_server/api/v1/kms/etsi004/get-key HTTP/1.1
Host: qkd_server
Content-Type: application/json
Accept: application/json
```

```
const inputBody = '{
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "index": 0,
  "metadata": {
    "size": 0,
    "buffer": ""
  }
}';
const headers = {
  'Content-Type':'application/json',
  'Accept':'application/json'
};

fetch('https://qkd_server/api/v1/kms/etsi004/get-key',
{
```

```javascript
  method: 'GET',
  body: inputBody,
  headers: headers
})
.then(function(res) {
    return res.json();
}).then(function(body) {
    console.log(body);
});
```

```ruby
require 'rest-client'
require 'json'

headers = {
  'Content-Type' => 'application/json',
  'Accept' => 'application/json'
}

result = RestClient.get 'https://qkd_server/api/v1/kms/etsi004/get-key',
  params: {
  }, headers: headers

p JSON.parse(result)
```

```python
import requests
headers = {
  'Content-Type': 'application/json',
  'Accept': 'application/json'
}

r = requests.get('https://qkd_server/api/v1/kms/etsi004/get-key', headers = headers)

print(r.json())
```

```php
<?php

require 'vendor/autoload.php';

$headers = array(
    'Content-Type' => 'application/json',
    'Accept' => 'application/json',
);

$client = new \GuzzleHttp\Client();

// Define array of request body.
$request_body = array();
```

```php
try {
    $response = $client->request('GET','https://qkd_server/api/v1/kms/etsi004/get-key', arra
        'headers' => $headers,
        'json' => $request_body,
       )
    );
    print_r($response->getBody()->getContents());
 }
 catch (\GuzzleHttp\Exception\BadResponseException $e) {
    // handle exception or api errors.
    print_r($e->getMessage());
 }

 // ...
```

```java
URL obj = new URL("https://qkd_server/api/v1/kms/etsi004/get-key");
HttpURLConnection con = (HttpURLConnection) obj.openConnection();
con.setRequestMethod("GET");
int responseCode = con.getResponseCode();
BufferedReader in = new BufferedReader(
    new InputStreamReader(con.getInputStream()));
String inputLine;
StringBuffer response = new StringBuffer();
while ((inputLine = in.readLine()) != null) {
    response.append(inputLine);
}
in.close();
System.out.println(response.toString());
```

```go
package main

import (
       "bytes"
       "net/http"
)

func main() {

    headers := map[string][]string{
        "Content-Type": []string{"application/json"},
        "Accept": []string{"application/json"},
    }

    data := bytes.NewBuffer([]byte{jsonReq})
    req, err := http.NewRequest("GET", "https://qkd_server/api/v1/kms/etsi004/get-key", data
    req.Header = headers
```

```
    client := &http.Client{}
    resp, err := client.Do(req)
    // ...
}
```

`GET /get-key`

start pushed key retrieval process

The `get-key` message is the starting point after which the QKD device can start pushing keys.

Body parameter

```
{
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20",
  "index": 0,
  "metadata": {
    "size": 0,
    "buffer": ""
  }
}
```

Parameters

Name	In	Type	Required	Description
body	body	object	true	none
» key_stream_id	body	string(uuid)	true	corresponding to the one given in `open_connect`
» index	body	integer(uint32)	false	increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.
» metadata	body	metadata	false	none
»» size	body	integer(uint32)	false	size of buffer in characters
»» buffer	body	string(json)	false	metadata in json format.

Example responses

205 Response

```json
{
  "status": 0,
  "metadata": {
    "size": 0,
    "buffer": ""
  }
}
```


Responses

Status	Meaning	Description	Schema
205	Reset Content	Successful registration	Inline
400	Bad Request	Client sent a malformed input	None
500	Internal Server Error	The server experienced an internal error	None

Response Schema

Status Code **205**

Name	Type	Required	Restrictions	Description
» status	status(uint32)	true	none	none

oneOf

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful connection, but peer not connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because insufficient key available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because peer application is not yet connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	No QKD connection available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN_CONNECT failed because the KSID is already in use

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	TIMEOUT_ERROR The call failed because the specified TIMEOUT

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because metadata field size insufficient. Returned Meta-data_size value holds minimum needed size of metadata

continued

Name	Type	Required	Restrictions	Description
» metadata	metadata	false	none	none
»» size	integer(uint32)	false	none	size of buffer in characters
»» buffer	string(json)	false	none	metadata in json format.

This operation does not require authentication

put_close

Code samples

```
# You can also use wget
curl -X PUT https://qkd_server/api/v1/kms/etsi004/close \
  -H 'Content-Type: application/json' \
  -H 'Accept: application/json'
```

```
PUT https://qkd_server/api/v1/kms/etsi004/close HTTP/1.1
Host: qkd_server
Content-Type: application/json
Accept: application/json
```

```
const inputBody = '{
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
}';
const headers = {
  'Content-Type':'application/json',
  'Accept':'application/json'
};

fetch('https://qkd_server/api/v1/kms/etsi004/close',
{
  method: 'PUT',
  body: inputBody,
  headers: headers
})
.then(function(res) {
    return res.json();
}).then(function(body) {
    console.log(body);
});
```

```
require 'rest-client'
require 'json'

headers = {
  'Content-Type' => 'application/json',
```

```ruby
  'Accept' => 'application/json'
}

result = RestClient.put 'https://qkd_server/api/v1/kms/etsi004/close',
  params: {
  }, headers: headers

p JSON.parse(result)
```

```python
import requests
headers = {
  'Content-Type': 'application/json',
  'Accept': 'application/json'
}

r = requests.put('https://qkd_server/api/v1/kms/etsi004/close', headers = headers)

print(r.json())
```

```php
<?php

require 'vendor/autoload.php';

$headers = array(
    'Content-Type' => 'application/json',
    'Accept' => 'application/json',
);

$client = new \GuzzleHttp\Client();

// Define array of request body.
$request_body = array();

try {
    $response = $client->request('PUT','https://qkd_server/api/v1/kms/etsi004/close', array(
        'headers' => $headers,
        'json' => $request_body,
       )
    );
    print_r($response->getBody()->getContents());
 }
 catch (\GuzzleHttp\Exception\BadResponseException $e) {
    // handle exception or api errors.
    print_r($e->getMessage());
 }

 // ...
```

```java
URL obj = new URL("https://qkd_server/api/v1/kms/etsi004/close");
HttpURLConnection con = (HttpURLConnection) obj.openConnection();
con.setRequestMethod("PUT");
int responseCode = con.getResponseCode();
BufferedReader in = new BufferedReader(
    new InputStreamReader(con.getInputStream()));
String inputLine;
StringBuffer response = new StringBuffer();
while ((inputLine = in.readLine()) != null) {
    response.append(inputLine);
}
in.close();
System.out.println(response.toString());
```

```go
package main

import (
       "bytes"
       "net/http"
)

func main() {

    headers := map[string][]string{
        "Content-Type": []string{"application/json"},
        "Accept": []string{"application/json"},
    }

    data := bytes.NewBuffer([]byte{jsonReq})
    req, err := http.NewRequest("PUT", "https://qkd_server/api/v1/kms/etsi004/close", data)
    req.Header = headers

    client := &http.Client{}
    resp, err := client.Do(req)
    // ...
}
```

`PUT /close`

close the key stream

The `close` message disconnects the KMS from the QKD device.

Body parameter

```json
{
  "key_stream_id": "4eb81a5c-031f-4d1f-881d-309bee44fc20"
}
```

Parameters

Name	In	Type	Required	Description
body	body	object	true	none
» key_stream_id	body	string(uuid)	true	corresponding to the one given in `open_connect`

Example responses

204 Response

```json
{
  "status": 0
}
```

Responses

Status	Meaning	Description	Schema
204	No Content	Key stream successfully closed	Inline
400	Bad Request	Client sent a malformed input	None
500	Internal Server Error	The server experienced an internal error	None

Response Schema

Status Code **204**

Name	Type	Required	Restrictions	Description
» status	status(uint32)	true	none	none

oneOf

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	Successful connection, but peer not connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because insufficient key available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because peer application is not yet connected

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	No QKD connection available

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN_CONNECT failed because the KSID is already in use

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	TIMEOUT_ERROR The call failed because the specified TIMEOUT

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred

xor

Name	Type	Required	Restrictions	Description
»» *anonymous*	integer	false	none	GET_KEY failed because metadata field size insufficient. Returned Metadata_size value holds minimum needed size of metadata

This operation does not require authentication

Schemas

qos

```
{
  "key_chunk_size": 32,
  "max_bps": 128,
  "min_bps": 64,
  "jitter": 1,
  "priority": 3,
  "timeout": 50,
  "ttl": 600,
  "metadata_mimetype": "application/json"
}
```

All values according to ETSI GS QKD 004 specification. If KMS does not want to specify a parameter, it is up to the choice of the QKD. Either the value is given as "null" or not present.

Properties

Name	Type	Required	Restrictions	Description
key_chunk_size	integer(uint32)	false	none	length of key chunks in Byte.
max_bps	integer(uint32)	false	none	in bit per second
min_bps	integer(uint32)	false	none	in bit per second
jitter	integer(uint32)	false	none	in bit per second
priority	integer(uint32)	false	none	Meaning up to implementation.
timeout	integer(uint32)	false	none	Timeout in msec after which the call will be aborted, returning an error.

Name	Type	Required	Restrictions	Description
ttl	integer(uint32)	false	none	Time in seconds after which the keys for this key stream ID shall be erased from the kms key store.
metadata_mimetype	string(json)	false	none	MIME type of the metadata of the `get` request

metadata


```json
{
  "size": 0,
  "buffer": ""
}
```


Properties

Name	Type	Required	Restrictions	Description
size	integer(uint32)	false	none	size of buffer in characters
buffer	string(json)	false	none	metadata in json format.

status

```
0
```

Properties

Name	Type	Required	Restrictions	Description
anonymous	integer(uint32)	false	none	none

oneOf

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	Successful

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	Successful connection, but peer not connected

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	GET_KEY failed because insufficient key available

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	GET_KEY failed because peer application is not yet connected

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	No QKD connection available

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	OPEN_CONNECT failed because the KSID is already in use

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	TIMEOUT_ERROR The call failed because the specified TIMEOUT

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	OPEN failed because requested QoS settings could not be met, counter proposal included in return has occurred

xor

Name	Type	Required	Restrictions	Description
anonymous	integer	false	none	GET_KEY failed because metadata field size insufficient. Returned Metadata_size value holds minimum needed size of metadata